Siraga







SIRAGA'S OFFER

catalogue

Cylinder handling & palletizing



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- Pallets p.9

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New!



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Pictures not contractual

Thanks to you, Siraga moves ahead !

Dear Customers,



Many of you have taken the time to answer our «Customer Satisfaction Questionnaire» and we warmly thank you for your time & high quality of your answers. Thanks to you, Siraga moves ahead! Indeed, who else could better judge the quality of our equipment and services than you? Your remarks and expectations enable us to pursue, faithful to our commitments, our continuous improvement policy, started almost 10 years ago!

Further to the survey, we have the pleasure to present our catalogue, which we hope to be clear, customer-friendly and last but not least, complete! You will find the comprehensive Siraga offer: may it be filling or refurbishing equipment, pallets or cylinder vending machines, along with our traditional services such as the LPG Academy, spare parts, after sales and eventually, Engineering and consulting (studies…). To put it in a nutshell, our global expertise, in only one single support!

We also underlined our know-how and especially the philosophy of design of our equipment: strongly-built and tailor-made machines, in the frame of essential safety, environment-friendliness and consumer-protection legislation, i.e. in conformity with the Directives applicable to your activity: ATEX…

More than a regular information leaflet, we hope the catalogue will become a real working tool! Of course, all our teams remain at your disposal for any comments or inquiry you may have!

Éric BATISE
General Manager

To receive the summary of our « Customer Satisfaction Questionnaire », only one address: marketing@siraga.com

Design of our equipment
Safety of people & machines



Olivier Becq,
Technical Director, Siraga

In the design and manufacture of our equipment, within the European Union we are subject to a number of directives and regulations in terms of safety, health, environment and consumer protection. We strictly respect these technical obligations. The CE marking on our machines is proof of this.
We therefore advise you to ensure that these markings are displayed on your equipment to certify that they comply with these regulatory requirements, if they are applicable to you.

olivier.becq@siraga.com

Focus on some New Approach Directives applicable within the EU

- **Electro-magnetic compatibility** (2004/108/EC)
  E.g.: ULIS PLC
- **Machines** (98/37/CE, amendment 98/79/EC)
  All our equipment
- **Non-automatic weighing instruments** (90/384/EEC, amendment 93/68/EEC)
  E.g.: Check scale
- **Explosive atmospheres** (94/9/EC)
  E.g.: ULIS PLC, electrical power
- **Pressurised equipment** (97/23/EC) DESP/PED Directives
  E.g.: LPG distribution gear, regulation skid, etc.

For example, the "Pressurised equipment" directive involves compliance with very specific criteria, such as:

- *Selection of materials and processes*
- *Qualification of welders & operating modes*
- *Inspection by a specialised body*

Preview of our machine design policy

We manufacture all our machines to order, which means we can offer you equipments that are tailor-made to suit your needs.
Furthermore, the purpose of our machine design policy is to guarantee you maximum reliability and safety, without jeopardising the functionality and competitiveness of our machines.

Quality Policy

The deployment of a quality management system ISO 9001, V2000 in 2001 for all our business enables us to self-certify for the Safety and ATEX standards (the mechanical portion). Through our organisation and the implementation of procedures, we can assure you that the equipment we manufacture is compliant with the models tested and presented to obtain the "EC TYPE" certificate.

We use electrical and electronic components that are ATEX certified. Concerning SIRAGA parts, we are, as well as our subcontractors, audited annually by a testing laboratory – the LCIE – to renew our ATEX notification.

E.g.: ULIS Plc, certified zone 0

The recent harmonisation of European and international standards in terms of ATEX regulations, the IECEx*, allows us to extend our strategy to a world-wide scope. In 2010, our equipment will comply with this new international standard.

**International Electrotechnical Commission scheme for Certification to Standards relating to Equipment for use in Explosive Atmosphere.*

Risk analysis

Assisted by a dedicated team from our engineering and design department, we conduct risk analysis of our machines. The assessment of risk (frequency/gravity) involves a series of logical steps to systematically examine the dangerous phenomena associated with the machine. The risk assessment is followed, whenever needed, by reduction or elimination of the risk, via different means such as modifications to the machine design, or implementation of protective devices, for example.
The same team tests equipment, in particular for electrostatic load tests (EN 13 463-I standard). The results of these analyses are then sent to the relevant laboratory (e.g. LCIE, etc.).

Lastly, an auditing body such as Bureau Véritas may be commissioned by the customer to ensure that the health and safety requirements are being followed, during an inspection of the equipment in our workshop and/or on its own site.

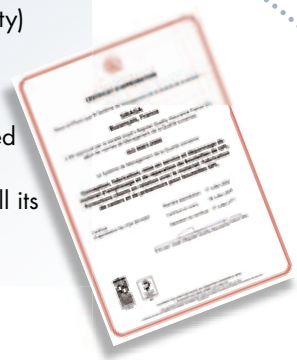
At your service for compliance and safety of your machines:

Manufacturer: Siraga

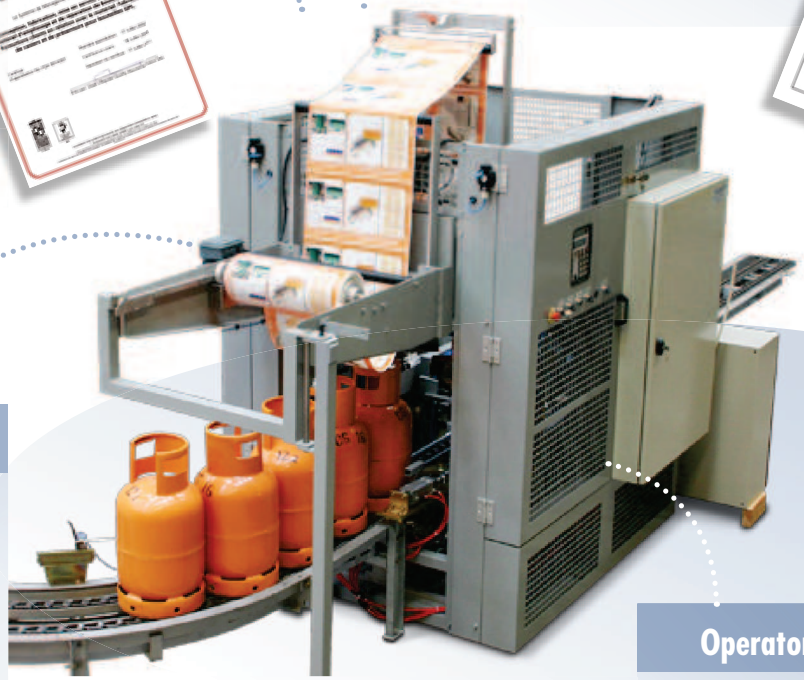
- Design
- Risk analysis (Atex/safety)
- Manufacture
- Tests
- Siraga has been certified ISO 9001, V2000 by LRQA since 2001, for all its business

Testing Laboratories:
E.g.: LCIE, Ineris, etc.

- Certification of electrical and electronic equipment
- ATEX Directive: e.g.: ULIS Plc
- Annual audit of Siraga







Auditing body:
E.g.: Bureau Véritas, Apave…

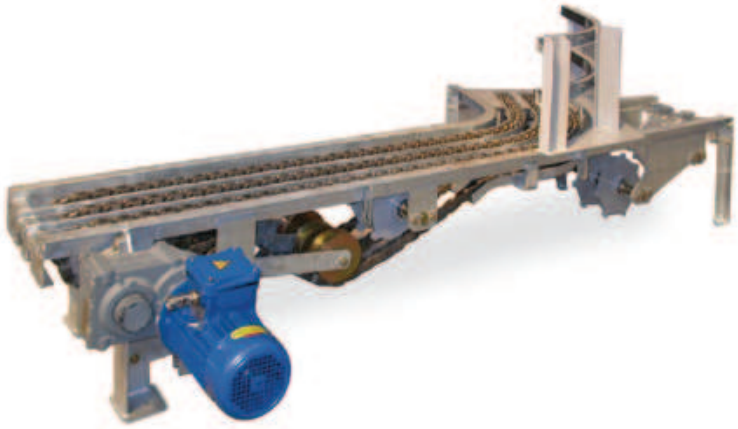
- Commissioned by the operator for checking compliance with health and safety requirements Imposed by the applicable directive: e.g.: Machines (98/37/EC, amendment 98/79/EC)
- Inspection at manufacturer site
- Inspection at operator site

Operator: Customer

- It is responsible for ensuring that all equipment installed on its site meets EC requirements

CE and Ex marking = visual proof of compliance with applicable directives

We therefore advise you to ensure that these markings are displayed on your equipment to certify that they comply with these regulatory requirements, if they are applicable to you.



Cylinder conveyor with lubricated or dry chain

- Chain speed from 6 to 35 meters/ minute
- Fully galvanized frame

Electrical driving unit for chain conveyor system with direct shaft mounting



Roller conveyor

- Metallic frame and galvanized rollers

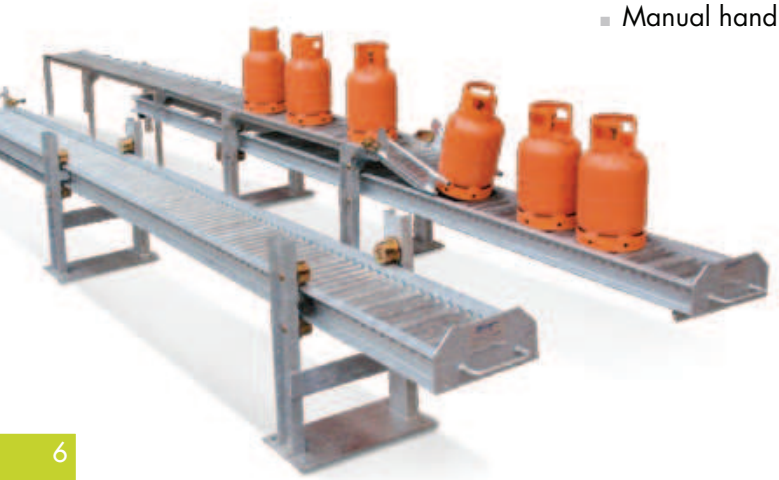


Telescopic chain conveyor

- Designed to load and unload un-palletized cylinders

Telescopic conveyor (with rollers)

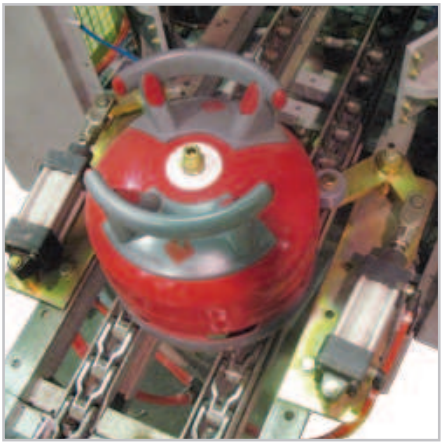
- Designed to load and unload un-palletized cylinders
- Manual handling

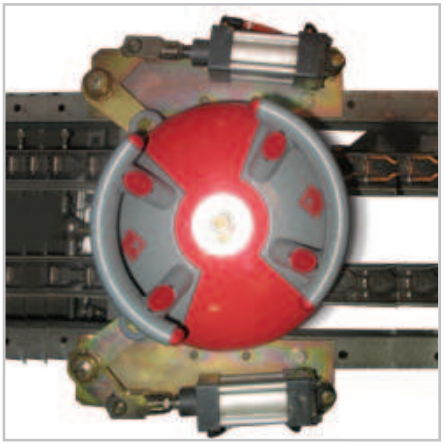




Over-head conveyor

- Automatic hooking and unhooking

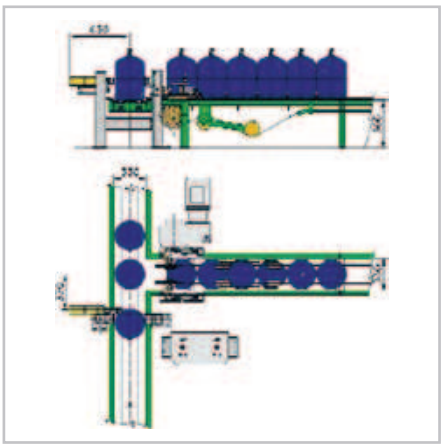




Cylinder stop unit
- Installed on chain conveyor



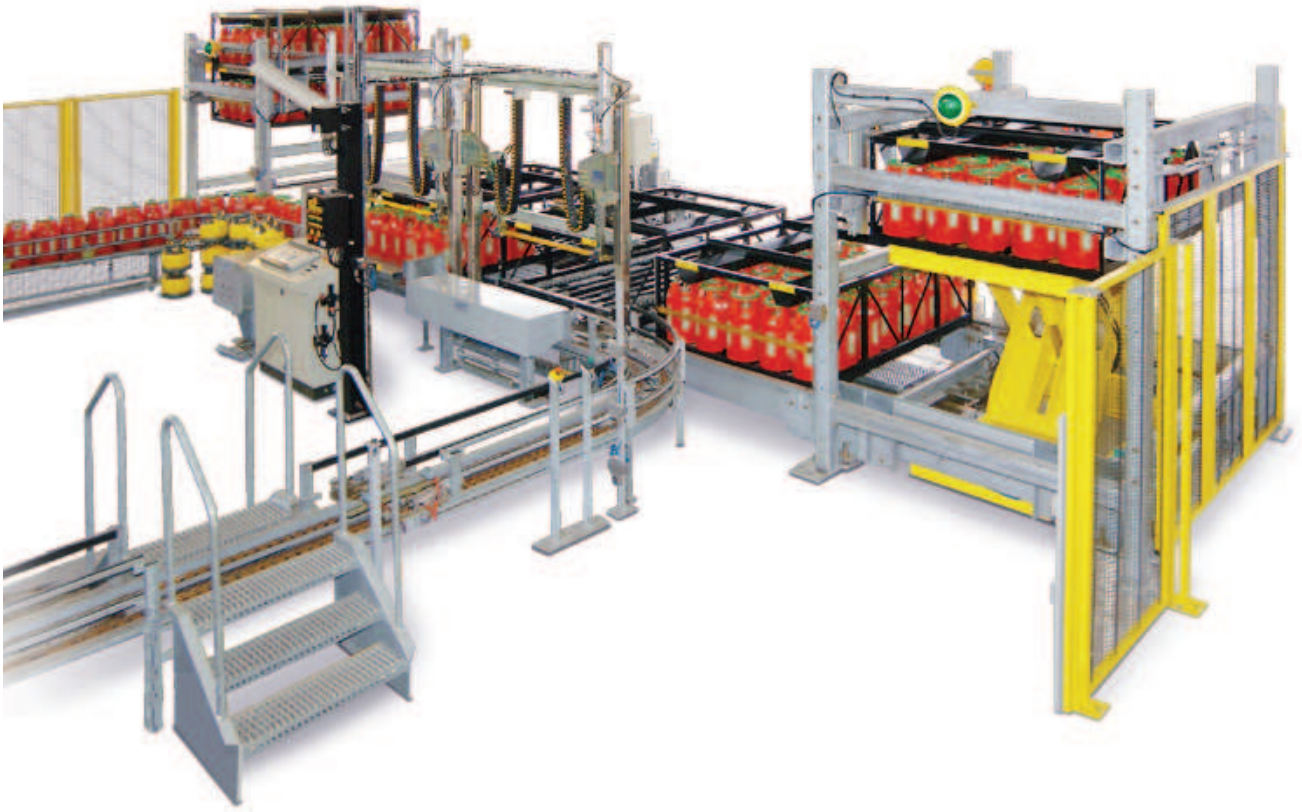
Cylinder stop unit
- For large-capacity cylinders



Automatic cylinder re-integration on main line system



Cylinder sorting device



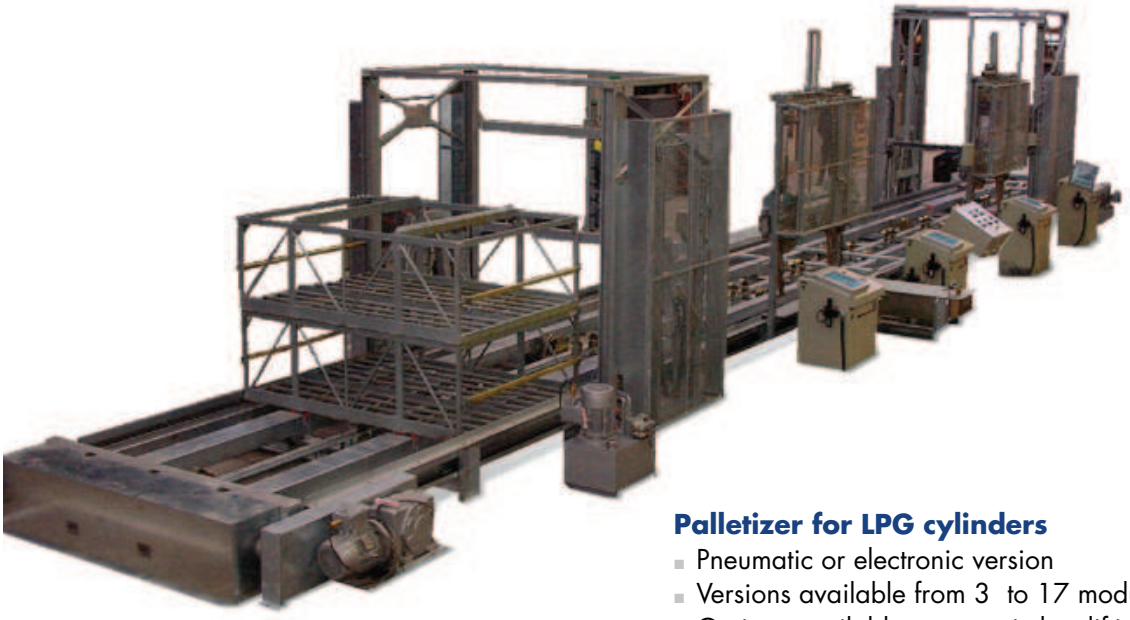
HYDRAULIS®

Electronic palletizer for LPG cylinders with hydraulic motorization

- Capacity up to 2,400 cylinders per hour
- Fully automatic
- Fully galvanized frame
- All types of pallets, even damaged ones







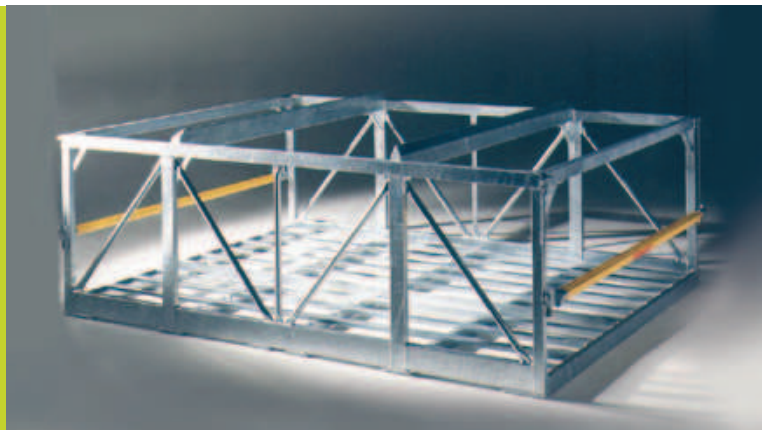
Palletizer for LPG cylinders

- Pneumatic or electronic version
- Versions available from 3 to 17 modules
- Options available: automatic bar lifting device – pallet stacking unit – pallet unstacking unit – pallet removing post of damaged pallets
- Also available for large-capacity cylinders (from 20 to 45 kg)

PALLETS & RACKS

Pallets for LPG cylinders

- Fiber glass closing bar
- Galvanised frame
- Possibility to supply pallets in kits (shipment by 80 or 125 kits, in 20' or 40' container)



Filling

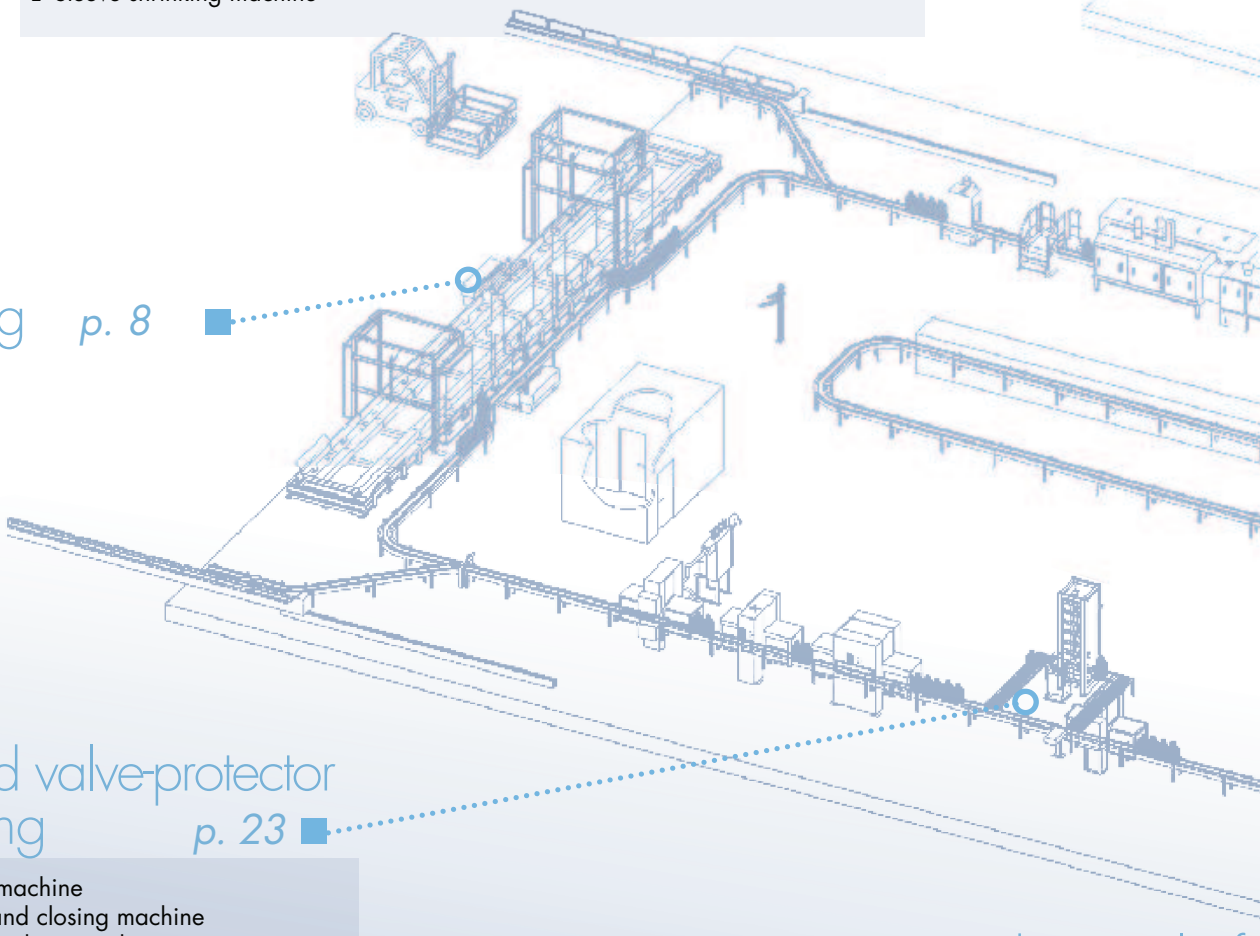
Capping & sleeving p. 26

Capping machine p. 26
- Semi-automatic capping unit for camping cylinders
- Automatic capping unit for vertical valves
- Pneumatic or hydraulic capping clamp

Sleeve shrinking machine p. 27
- Thermo-shrinkable sleeve laying machine
- Sleeve shrinking machine
- Sleeve shrinking machine for various cylinder sizes
- Sleeve shrinking machine (manual version)
- Streamer laying machine
- Cylinder Label laying machine

NEW! **Streamer removing machine**

Palletizing p. 8



Valve and valve-protector processing p. 23

- Valve orienting machine
- Valve opening and closing machine
- Valve closing checking machine

Valve screwing and unscrewing p. 24

NEW! **Valve screwing / unscrewing machine with torque control & display**
- Valve screwing and unscrewing post
- Hydraulic valve screwing and unscrewing machine
- Side-entry/top-entry valve changing unit

Screwing and unscrewing of valve protectors p. 25
- Carousel to automatically remove the metallic cap from cylinder

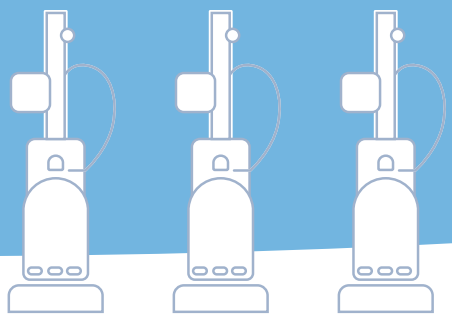
Security and control of

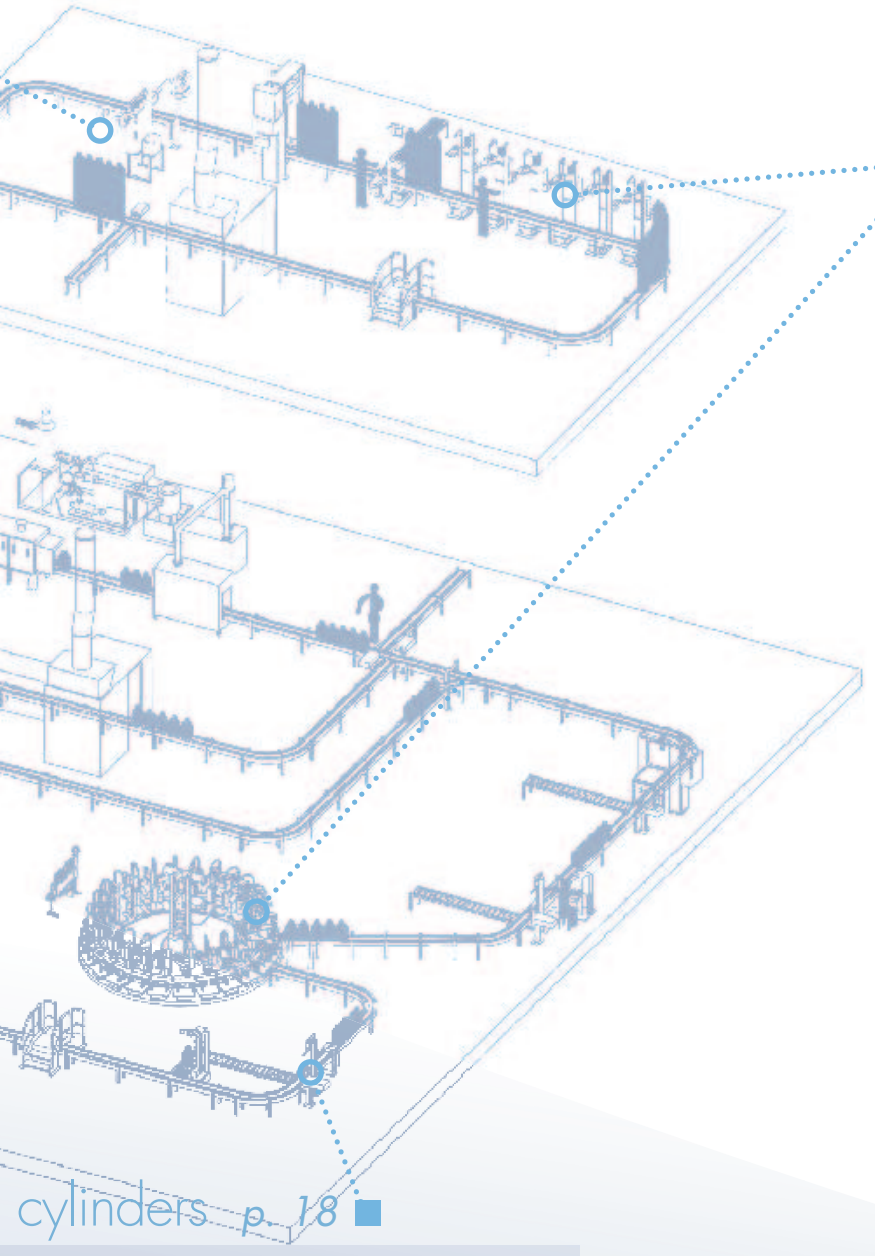
Weight check scale p. 18

NEW! **Dynamic check scale**
- In line weight check scale
- Electronic weight check scale

Leak detector p. 19

NEW! **Valve leak and gasket detector**
- Electronic leak detector
- Electronic leak detector for camping cylinders
- Electronic leak detector for cylinders of various sizes
- Electronic leak detector (double detector version)
- Electronic leak detector (version with valve opening and closing)





Filling *p. 12*

Cylinder admission unit on carousel p. 12
- In line system
- Tangential system
- System for large capacity cylinders
- In line system with two arms working alternatly

LPG bottling on carousel p.12
Bottling by electronic filling scales p.14
Bottling by mass flowmeter p.14
Stationary bottling or in line p.15
- Electronic stationnary filling scales for cylinders from 2 to 10 kg
- Electronic stationnary filling scales for cylinders from 10 to 100 kg
- In line electronic filling scales for cylinders from 2kg up to 100 kg
- Automatic in line filling system
- Multiple filling scales

Filling skid p.16
Filling container p.16
Weight resetting scale p.16
NEW! **Weight resetting & emptying scale**
Software for production management p.17
Global filling line management p.17

cylinders *p. 18*

- Electronic leak detector (manual version)
- A test cylinder to calibrate leak detectors

Valve gasket detector p. 21
Leak detection check p. 21
- In-line leak detection with water bath
- Tilting leak detection with water bath
- Machine to check the valve flow limitor presence (only for valves fitted with flow limitor)
- Automatic detection of damaged collar ring
- Multi-control machine for large-capacity cylinders

Annex equipment *p. 29*

LPG pressure regulation p. 29
LPG pressure regulation skid
LPG pumps and compressors p. 29
LPG transfer group
LPG pumping group

Cylinder admission unit on carousel

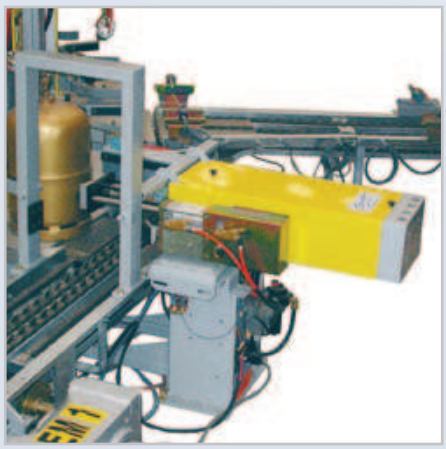


In line system
- Capacity up to 1,500 cylinders/ hour



In line system
- Cylinders carried by the conveyor
- Capacity up to 1,500 cylinders/hour

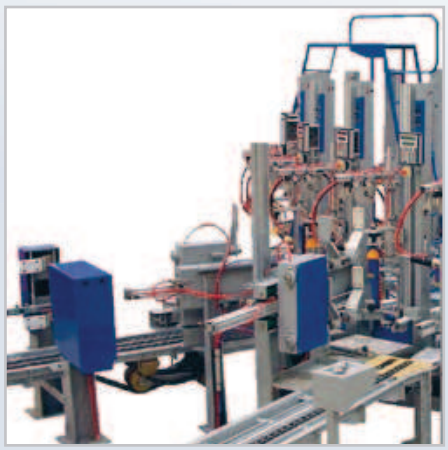




Tangential system
- Capacity up to 1,500 cylinders/hour

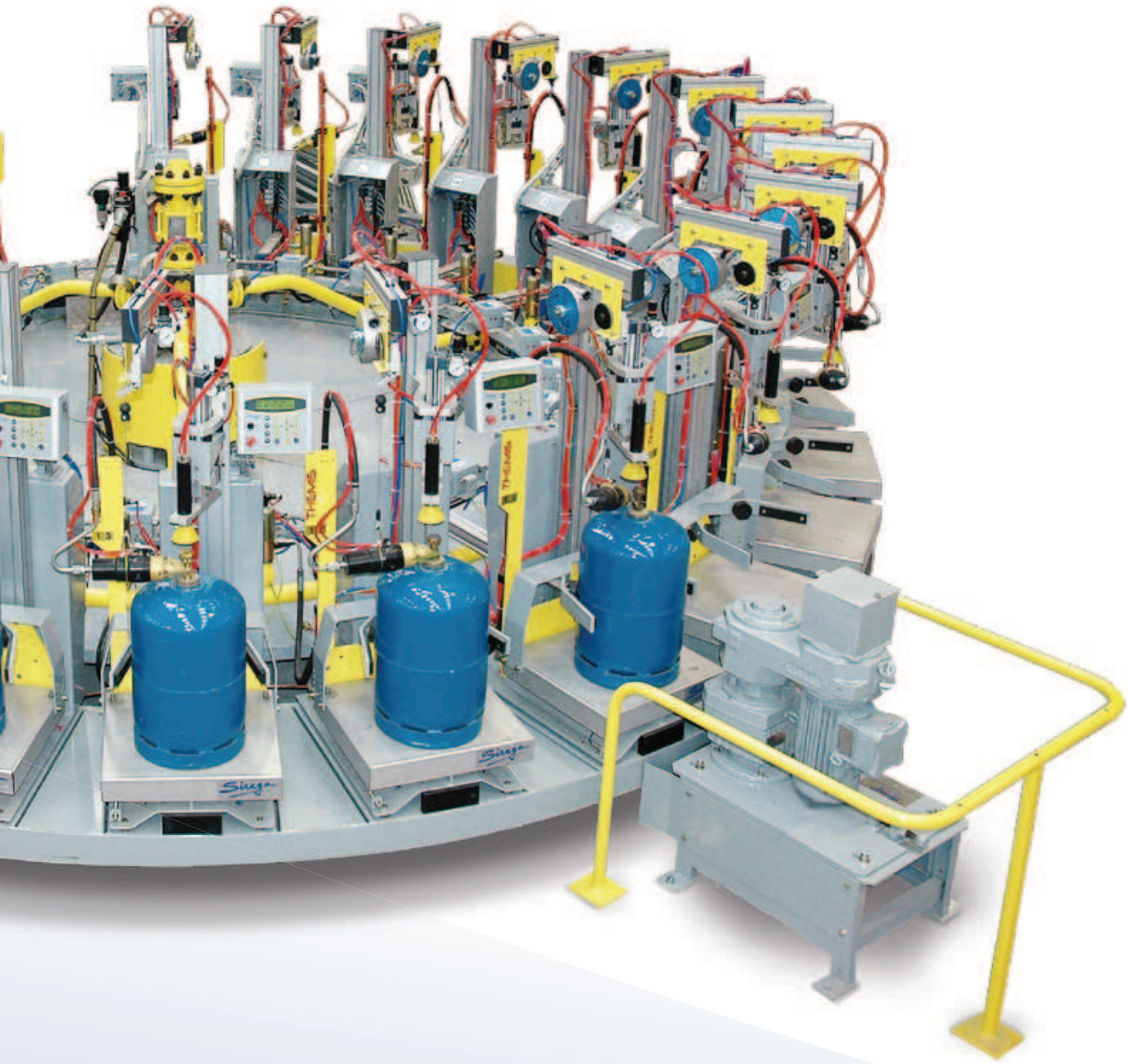


In line system with two arms working alternatly
- Capacity up to 2,000 cylinders/hour



System for large capacity cylinders
- Together with a cylinder extraction device at the carousel outlet

LPG bottling carousel with mass flow meter or with load cells



- All sizes of cylinders (from small to high capacity)
- All types of materials (steel, composit, ...)
- All types of valves
- Available in 8, 12, 18, 24, 30, 36 posts
- Capacity from 300 to 1,800 cylinders/hour
- One free post can be reserved for maintenance inside carousel

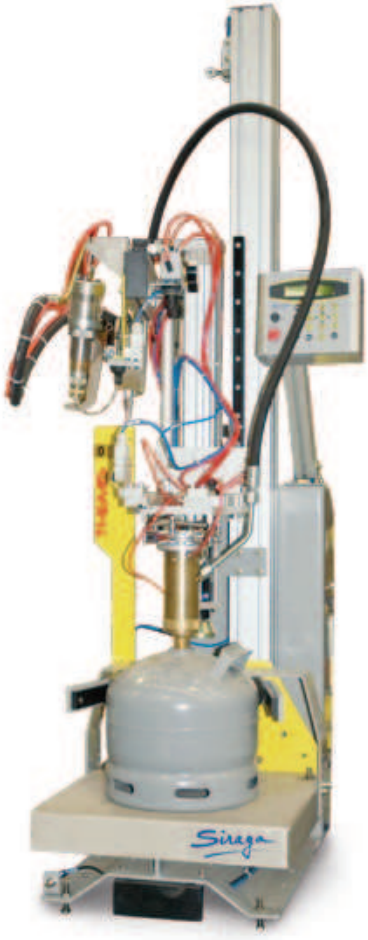
Electronic filling scales







Manual or automatic connection for all types of cylinders and valves





- Side and top-entry valve, with automatic valve opening and closing

Massflow meter



- Manual or automatic connection for all types of cylinders and valves

Stationary or in line filling



Electronic stationary filling scales for cylinders from 2 to 10 kg

- For all types of valves
- Multi connections available for connecting different valves on same scales
- 300 kg capacity load cell
- Very strong and robust design

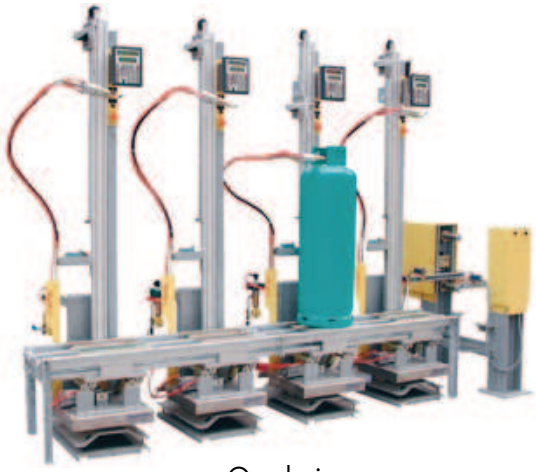




Electronic stationary filling scales for cylinders from 10 to 100 kg

- For all types of valves
- Multi connections available for connecting different valves on same scales
- 300 kg capacity load cell
- Very strong and robust design

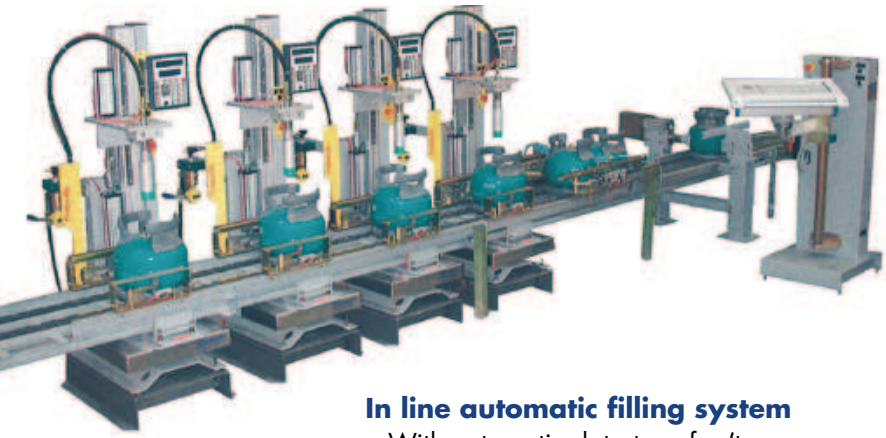
In line electronic filling scales for cylinders from 2 kg to 100 kg



On chain conveyor



On roller conveyor



In line automatic filling system

- With automatic data transfer (tare weight – filling result) from the tabulator to the filling scale and finally to the weight control scale



Multiple filling scales

- For cylinders, from 3 to 6 kg, equipped with Camping Gaz valves
- For cylinders, up to 15 kg, equipped with side-entry valves

Filling skid



Filling solution adapted to low production

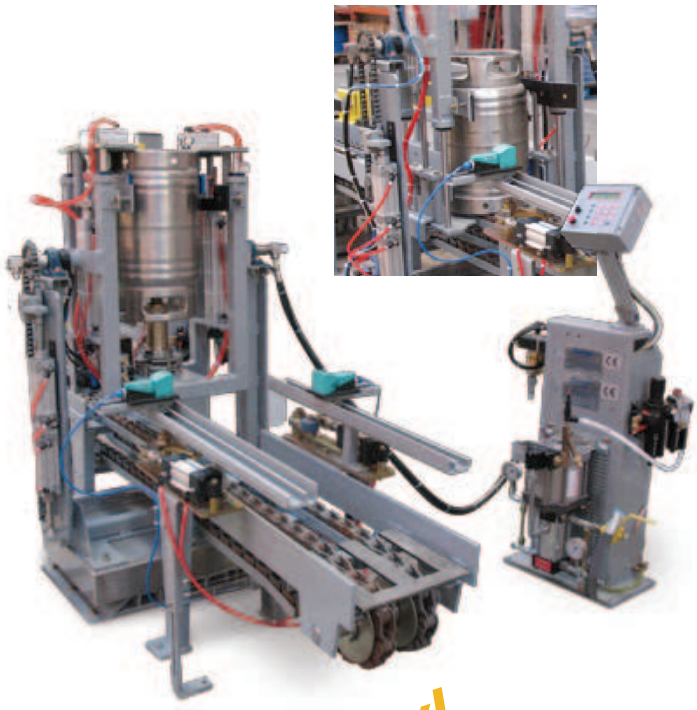
- Filling scale
- Control scale
- Leak control device
- Production from 30 to 250 cylinders per hour

Filling container



- Made from a maritime container, this filling hall is fully autonomous and meets safety standards

Weight resetting scale



New!

LPG cylinder weight resetting scale

- In line
- Electronic, manual or automatic version

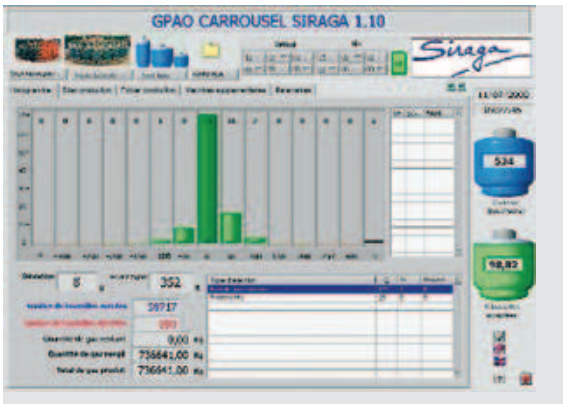


- This equipment corrects the weight of under/over filled cylinders
- In automatic version, gas adding or removing is automatic after entering the nominal weight

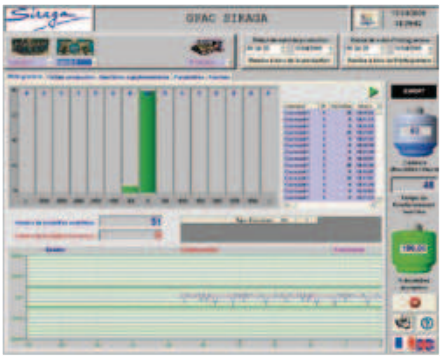
Software for production management

SIRAGA has developed a specific software dedicated to the treatment of data from carousel or in line filling system – all data cylinder by cylinder are stored on computer and allow daily, weekly, monthly or annually easy calculation.
Some data available: quantity of remaining gas in cylinders – quantity of filled gas – quantity of gas sold – filling accuracy for each cylinder.

When this software is linked to an automatic infra red tag reader - to read matricial code on cylinder or on the valve – it is possible to identify data of each cylinder with the serial number of the cylinder.

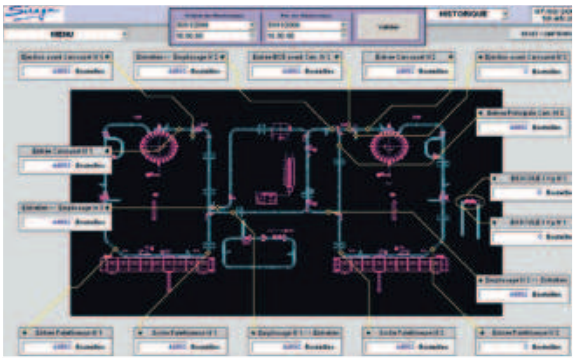


Global filling line management

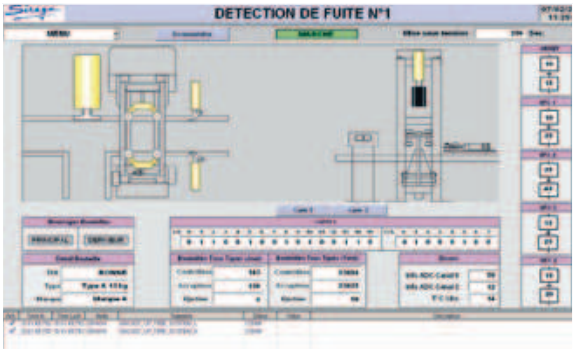


Carrousel supervision

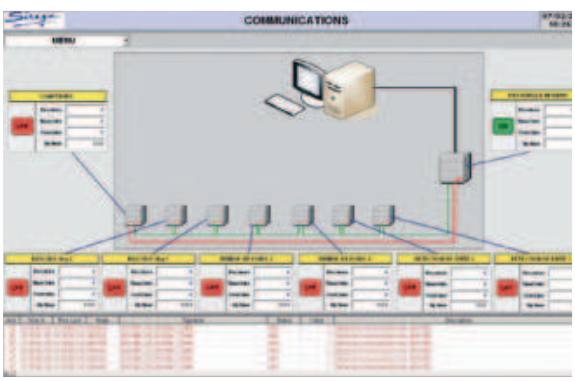
Considering the importance that computer control has taken on in the world of industry, the production management in a filling hall has become essential. The generalisation of machine control by an independent PLC enables us to extend the control of the carousel and the check scale to all the machines controlled by our ULIS PLC. (Leak detection or any other machine upstream or downstream the filling operation).



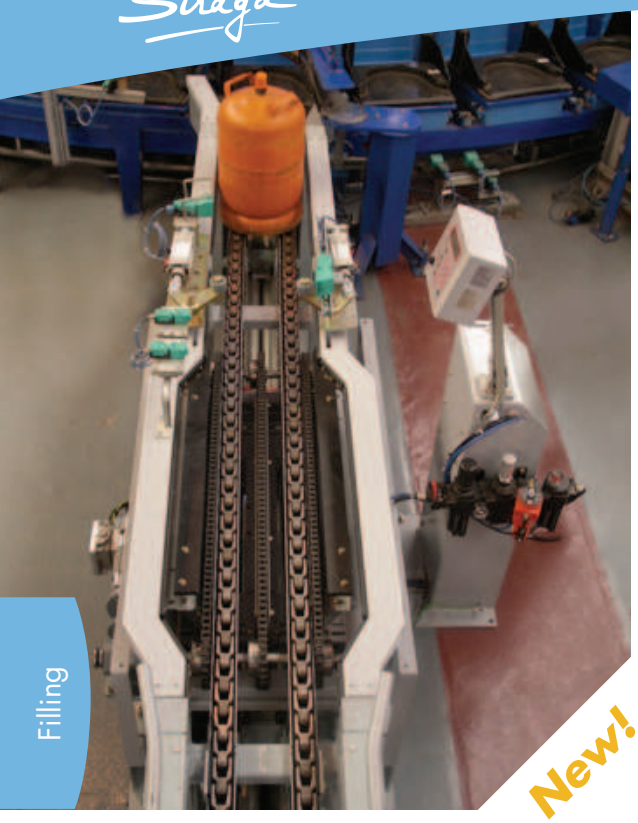
Metering



Machine condition



Communication



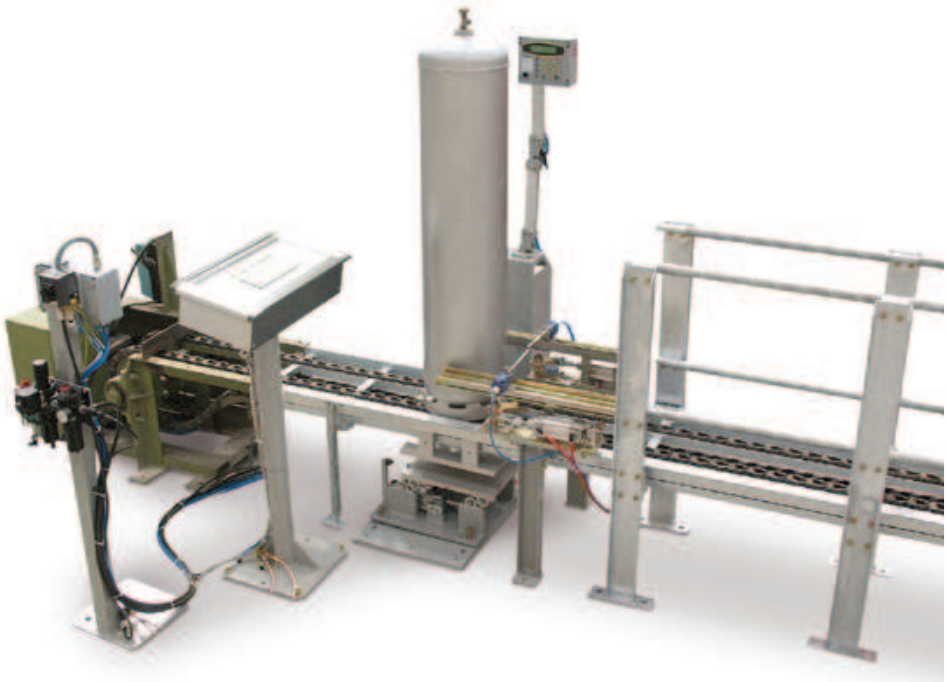
Weight check scale

New!

Dynamic check scale
- Capacity: above 1,600 cylinders/hour
- Accuracy: 50 g



In line weight check scale
- Single or double version
- Installed on chain conveyors, it allows the automatic weight checking up to 2,000 cylinders per hour



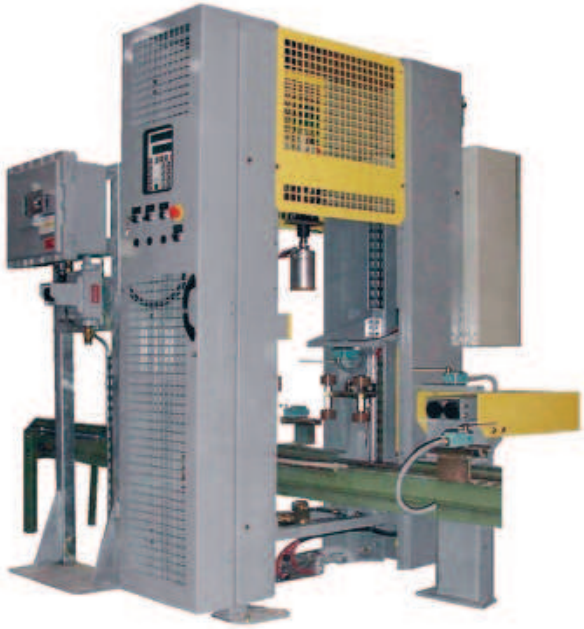
Weight check scale
- For large capacity cylinders

Leak detector



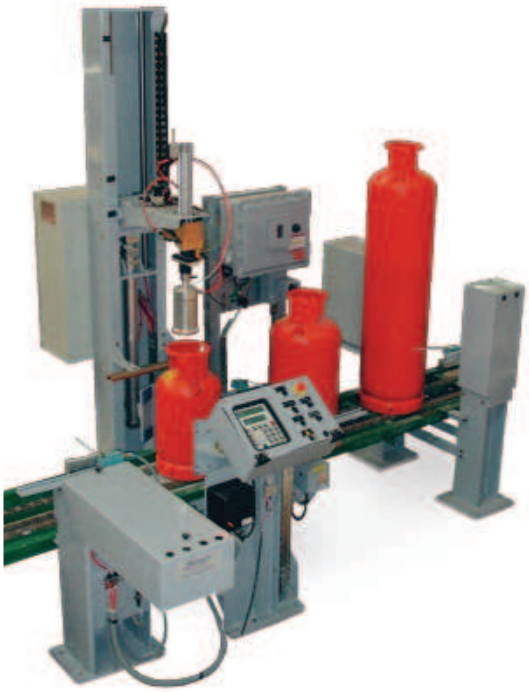
Valve leak & gasket detector

- One or two head machine
- Capacity for one head: 1,100 cylinders/hour
- Two heads: above 1,500 cylinders/hour
- Leak detection: 0.5 grams /hour
- Valve thread leak: 2 grams per hour
- Detection principle with double control: by electronic pressure sensor & infrared analysis



Electronic leak detector

- Capacity up to 1,200 cylinders per hour
- Detection up t 1.5 grams per hour



Electronic leak detector for various size cylinders

- Detection up to1.5 grams per hour



Electronic leak detector for camping cylinders

- Capacity up to 1,500 cylinders per hour
- Detection up to 1.5 grams per hour



Electronic leak detector
Double version detector
- Capacity: up to 1,800 cylinders per hour
- Detection: 1.5 grams per hour



Electronic leak detector
Version with automatic valve opening and closing
- Capacity: 1,200 cylinders per hour
- Detection: 1.5 grams per hour

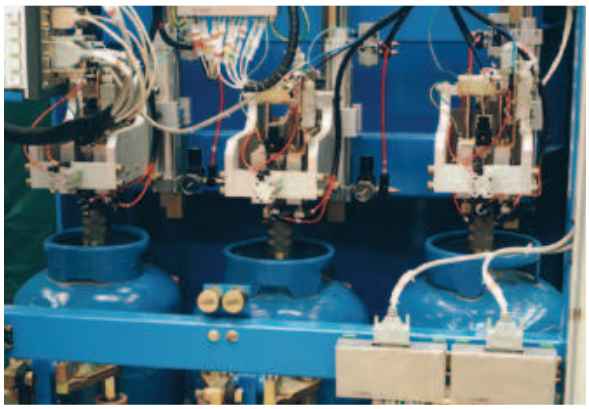


Electronic leak detector
Manual version
- Detection: 1.5 grams per hour for valves
- 3 grams per hour for side-entry valves

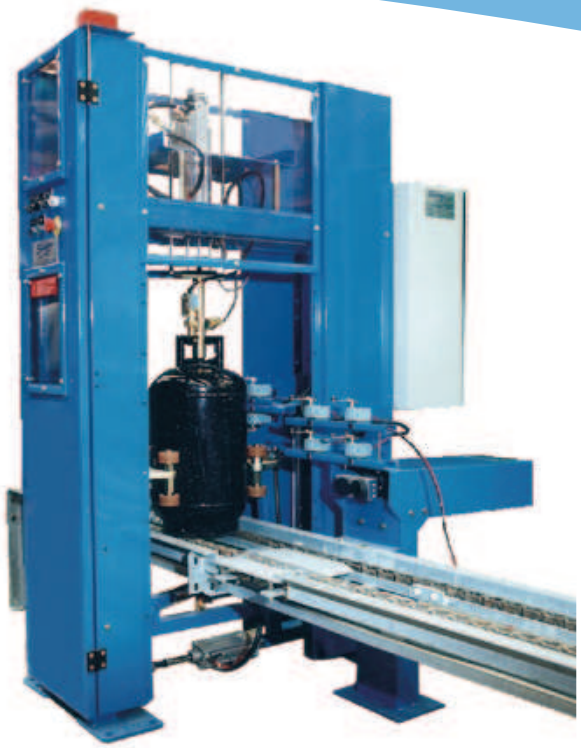


Test cylinder to calibrate leak detectors
- Temperature and pressure balanced
- No adjustment necessary
- Pre-calibrated upon request at 1.5g/h, 3 g/h and 5 g/h

Valve gasket detector



Valve gasket detector
Double or triple version
- Capacity up to 1,800 cylinders per hour



Valve gasket detector
- Capacity up to 1,200 cylinders per hour

Leak detection check



In-line leak detection with water bath
- Cylinders are driven by chain conveyor
- Model presented is for 13 kg cylinders



In-line leak detection with water bath
- Cylinders are driven by chain conveyor
- Model presented is for camping cylinders



Tilting leak detector support with water
- For water immersion with low production
- Manual or pneumatic functioning



Machine to check the valve flow limitor presence (only for valves fitted with flow limitor)

- Capacity: 1,200 cylinders
- For side entry or top entry valve



Automatic detection of damaged collar ring

- Sorts out unacceptable cylinders due to damaged collar ring
- Electronic version
- Capacity: 1,200 cylinders/hour = 1 head
  1,800 cylinders/hour = 2 heads



Multi-control machine for large capacity cylinders

- 3 functions in the same machine
- Electronic leak detector
- Overfilling detector.(liquid gas level)
- Control of the torque
- Capacity up to 250 cylinders per hour

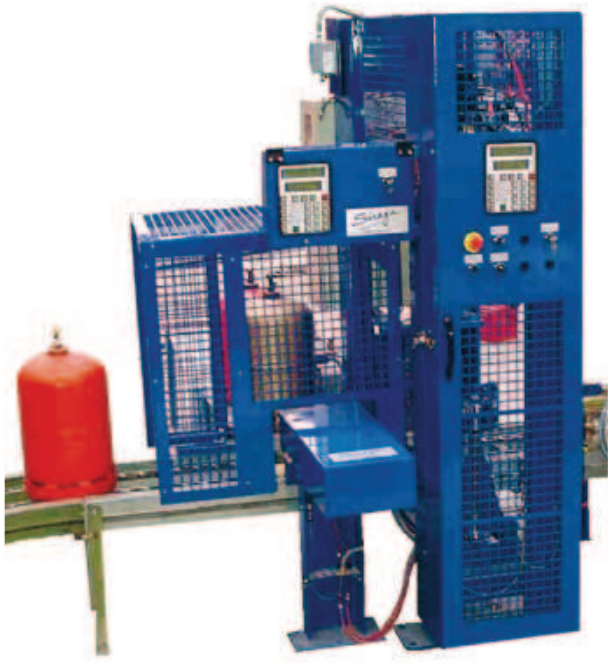




Valve orienting machine

- Single orientation version or with valve opening before introduction on carousel
- Electronic version
- Capacity up to 1,200 cylinders





Valve opening and closing

- Capacity: 1,200 cylinders/hour = 1 head
  1,800 cylinders/hour = 2 heads

Machine to check the torque

- Checking of the torque & of the number of rotations required to open & close the valve
- Electronic version
- Capacity up to 1,200 cylinders per hour

Valve screwing and unscrewing







Valve screwing / unscrewing machine with torque control & display

- Stationary machine
- Capacity from 100 to 120 cylinders/hour
- With or without collar ring
- Small & large capacity cylinders

Valve screwing and unscrewing post

- For all kinds of cylinders
- With manual or pneumatic screw
- On conveyor or stationary



Side-entry/top-entry valve changing unit on filled cylinders

- The preliminary cylinder draining is not necessary!
- Semi-automatic functioning
- Electronic version
- Capacity up to 60 cylinders per hour



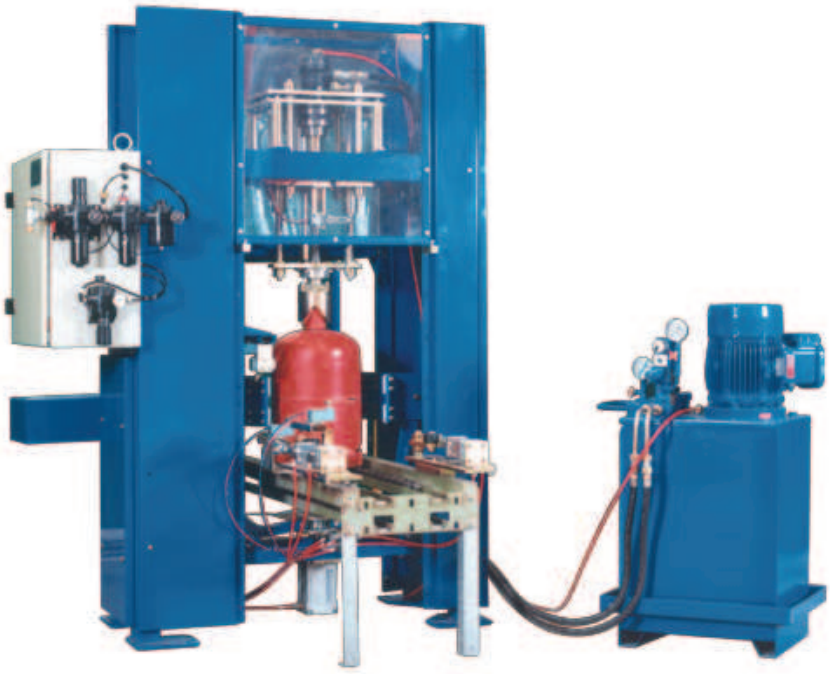


Hydraulic valve screwing and unscrewing machine on conveyor or stationary

- Torque adjustment according to the hydraulic pressure
- Bi-manual command for the operator
- 2 speeds: starting and finishing

Screwing and unscrewing of valve protectors





Screwing and unscrewing of valve protectors

- All these equipment are specific to certain countries where the valve is protected with a screwed metallic protector (ex. France…)

Machine to screw or unscrew automatically the valve metallic protector

- All types of cylinders
- Production capacity up to 1,200 cylinders/hour
- Torque control on screwing operation and unblocking security by unscrewing 1/4 back
- Automatic ejection for all cylinders with metallic protector blocked or impossible to screw/unscrew



Carousel to remove automatically the metallic protector from the cylinder

- All types of cylinders
- Production capacity 2,400 cylinders/hour

Capping machine



Semi-automatic capping unit for camping cylinders

- Production: up to 1,500 cylinders per hour



Automatic capping unit for top-entry valves

- Electronic version
- Production up to 1, 800 cylinders per hour





Pneumatic or hydraulic capping clamp

- According to the cap type

Sleeve laying & shrinking machine





Thermo-shrinkable sleeve laying machine

- Capacity up to 1,200 cylinders per hour

Sleeve shrinking machine

- Functioning with hot air
- Certified Eex IIB T2
- Capacity up to 1,200 cylinders per hour in simple version or 1,800 cylinders per hour double version, depending on the sleeve type & quality
- Retractable head







Sleeve shrinking machine Version for various cylinder sizes

- Functioning with hot air
- Certified Eex IIB T2

Sleeve shrinking machine Manual version

- Functioning with hot air
- Certificied Eex IIB T2

Sleeve shrinking machine

- Capacity:
  1,200 cylinders per hour with one head
  1,800 cylinders per hour with two heads
- Fixed head





Streamer laying machine
- Capacity up to 1,200 cylinders per hour



Label laying machine
Plastic or other label to inform the end-user
- Capacity from 1,200 up to 1,800 cylinders per hour

New!





Streamer removing machine
- Capacity up to 1,200 cylinders per hour

LPG pressure regulation

LPG pressure regulation skid

- Ensures LPG pressure regulation to one or more carousels or individual scales
- Regulation valve system or by discarding device



LPG pumps and compressors



LPG transfer group

- With LPG compressor





LPG pumping group

- Any flows

Refurbishment & maintenance

Gas purging *p. 34*

- Emptying support
- Cylinder tilting emptying support
- Emptying unit
- Inside washing machine
- Machine to remove residues
- Bung cleaning machine

Testing *p. 40*

Water testing ramp p. 40
- Testing rack
- Tilting testing rack
- Automatic testing rack
- Testing unit

Air testing ramp p. 41
- In line emptying
- Tilting leak testing

Palletizing *p. 8*

Tare marking *p. 43*

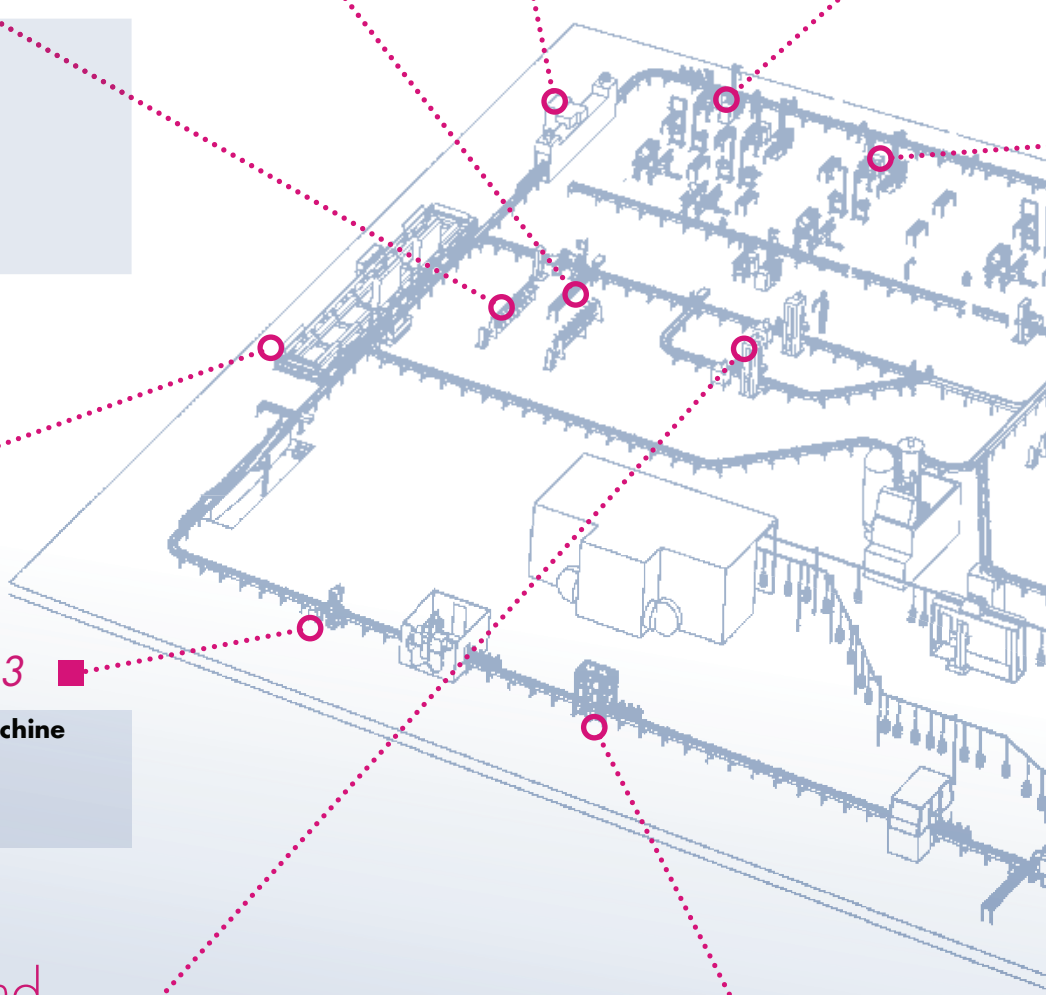
NEW! **Cylinder tare marking machine**
- Electronic cylinder taring scale
- Tare marking post

Valve screwing and unscrewing *p. 36*

NEW! **Valve screwing / unscrewing machine with torque control & display**
- Valve screwing and unscrewing post
- Hydraulic valve screwing and unscrewing machine
- Side-entry/top-entry valve changing unit on filled cylinder

Washing *p. 44*

- Washing tunnel
- High pressure washing tunnel
- Drying system
- Detergent spraying tunnel
- Pump skid for high pressure washing tunnel
- Water decantation unit



Gassing unit *p. 42*

- Gassing unit
- Gassing under vacuum unit
- Automatic gassing unit for LPG cylinder

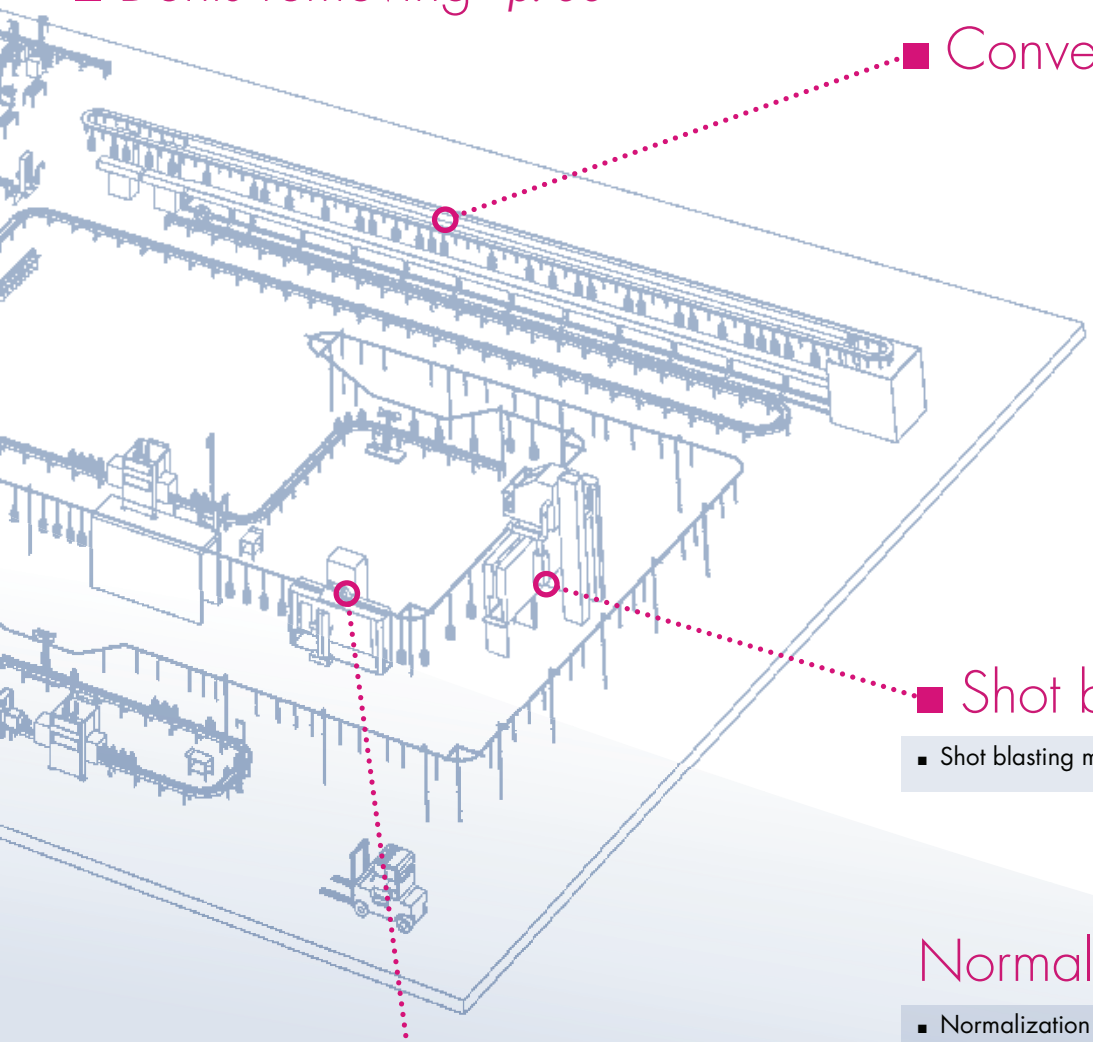


Foot-ring and shroud processing *p. 37*

Foot-ring and shroud straightening unit p. 37
- Foot-ring or shroud straightening unit

Foot-ring and shroud replacement unit p. 37
- Foot-ring cutting unit
- Grinding unit
- Welding unit

Dents removing *p. 38*



Conveyor *p. 6*

Destruction *p. 43*

- Machine to crush and pierce cylinders
- Machine to flatten cylinders
- Machine to pierce cylinders

Shot blasting *p. 38*

- Shot blasting machine

Normalization *p. 38*

- Normalization furnace

Metallization *p. 38*

Painting *p. 39*

- Painting booth for LPG filling plant
- Rotating device for manual painting
- Polymerization unit

Repairing of LPG cylinder



Foot-ring & valve protector welding *p. 37*



Dents removing *p. 38*



High pressure test *p. 40*

Shot blasting *p. 38*

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Normalizing *p. 38*

8

Foot-ring and valve protector removing *p. 37*

7

6

5



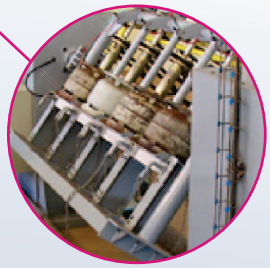
Grinding *p. 37*

4



Valve removing *p. 37*

3



Degassing with water *p. 34*

2

Emptying *p. 34* 1





Foot-ring & valve protector straightening *p. 37*



Metallization
p. 38



Polymerization
p. 39



Air filling

12

13

Painting
p. 39

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Valve screwing
p. 36

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Air purging

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Water testing
p. 40

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Gassing *p. 42*

Tare weighing
p. 43

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21



Marking *p. 43*





Large capacity cylinder emptying
- System allowing the operator to tilt cylinders easily



Cylinder emptying support
- All types of cylinders with all types of valves

Pneumatic draining pump

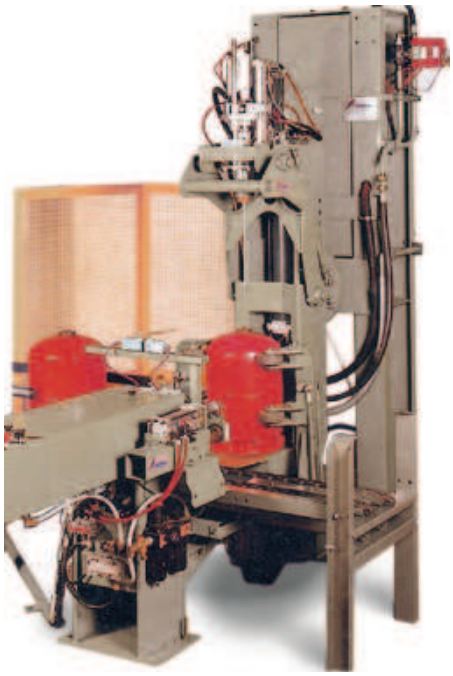




Cylinder tilting emptying support for large capacity cylinder
- Multiple post

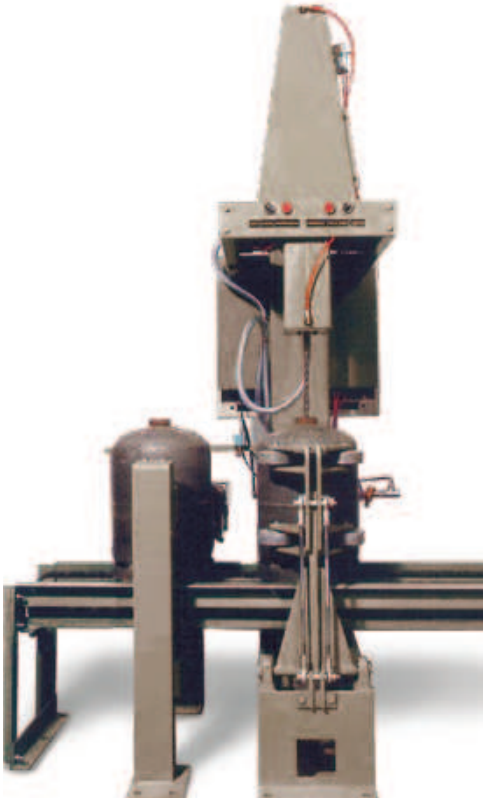


Emptying unit
- Functioning by LPG compressor with tanks to receive gas sucked from cylinders and with discharge to the storage
- Fully automatic



Inside cylinder washing machine

- Capacity up to 120 cylinders per hour
- High pressure cleaner 65/90 bars
- Automatic version



Machine to remove residues

- Residual water suction after inside washing
- Capacity up to 300 cylinders/hour
- Automatic version



Removing residual unit

- Residual water suction after inside washing
- Capacity up to 300 cylinders/hour
- Manual version



Bung cleaning machine

- Inside & outside ring Cleaning

VALVE SCREWING AND UNSCREWING

Siraga









Valve screwing and unscrewing post

- For all kind of cylinders
- With manual or pneumatic screw
- On conveyor or stationary

Valve screwing / unscrewing machine with torque control & display

- Stationary machine
- Capacity from 100 to 120 cylinders/hour
- With or without collar ring.
- Small & large capacity cylinders



Side-entry/top-entry valve changing unit on filled cylinders

- The preliminary cylinder draining is not necessary!
- Semi-automatic functioning
- Electronic version
- Capacity up to 60 cylinders per hour





Hydraulic valve screwing and unscrewing machine on conveyor or stationary

- Torque adjustment according to the hydraulic pressure
- Bi-manual command for the operator
- 2 speeds: starting and finishing

Foot-ring and collar straightening unit



Foot-ring or shroud straightening unit

- All types of cylinders
- Vertical or horizontal machines
- 2 work sides on horizontal cylinders, 2 types of foot-ring or valve protector on the same machine





Foot-ring and collar replacement unit



Foot-ring cutting post

- All types of cylinders
- By plasma cutting or chiselling



Grinding post

- All types of cylinders



Welding post

- All types of cylinders
- Manual, semi-automatic or automatic version

DENTS REMOVING POST



NORMALIZATION



Normalization furnace
- All types of cylinders
- Capacity from 50 to 600 cylinders/hour

SHOT BLASTING



Shot blasting machine
- All types of cylinders
- Horizontal or vertical on overhead conveyor

METALLIZATION



Zinc spraying unit
- Manual or automatic





Painting booth for filling plant, refurbishing plant or manufacturing plant

- Painting system up to 1,800 cylinders per hour
- Water curtain or dry painting booth – installation on chain or overhead conveyor
- Protection system for the valve
- Classical liquid or electrostatic paint
- possibility to install an automatic painting booth on chain

Cylinder rotating device with manual painting

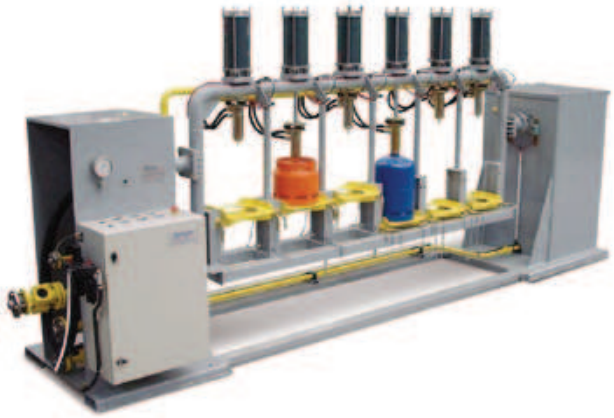


Polymerization unit

- All types of cylinders



Water testing ramp



Testing rack

- 1, 2, 3, 6, or 10 testing posts
- All kinds of cylinders
- Manual or automatic rotation.
- Possibility of automatic leak detection with electronic management



Tilting testing rack

- Version for different size of cylinders





Testing unit

- To connect to testing ramp with low and high pressure water pump
- With galvanized water tanks
- With compressed air tanks
- With simple or multiple decantation system

Tilting testing rack

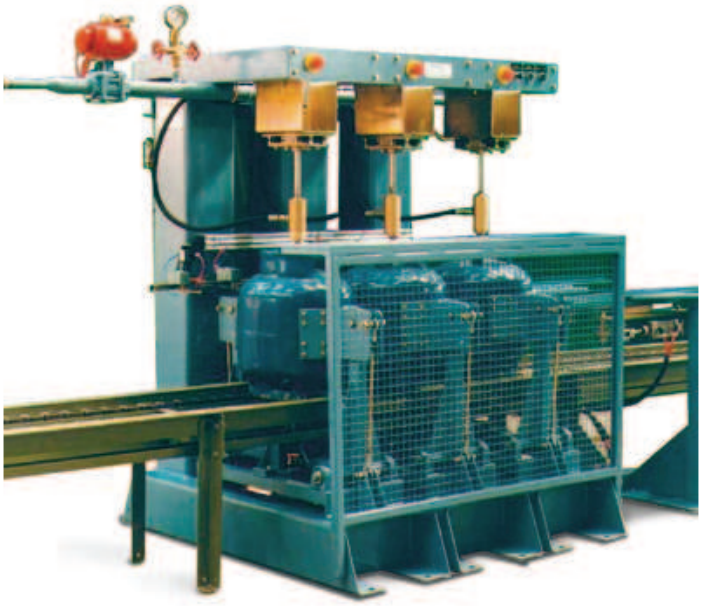
- Version for large capacity cylinders



Automatic testing rack for 26 litres cylinder

- In line functioning on chain conveyor
- Electronic management

Air testing ramp



In line emptying

- On chain conveyor



Tilting leak testing

- In water bath
- Manual or pneumatic tilting
- Number of posts according to production

GASSING UNIT

Siraga





Gassing under vacuum unit

- 1 up to 4 posts under a vacuum pump
- Stationary or on chain conveyor

Gassing unit

- Connected to the network or supplied from a reversed cylinder



Automatic gassing unit for LPG cylinder

- This machine releases automatically a certain quantity of liquid gas in cylinder free from their valve. This operation removes the air inside the cylinder

TARING



New!

Machine to mark valve protector

- Production: from 350 to 400 cyl/ hour for a chain speed at 8 meters/ minute



Electronic cylinder taring scale

- Stationary or on chain or roller conveyor



Cylinder tare stamping unit

CYLINDER DESTRUCTION



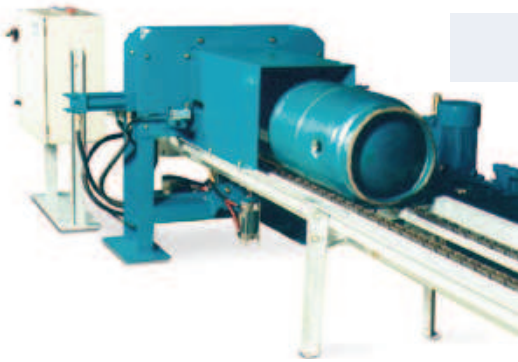
Cylinder crushing and piercing unit

- Pierces and crushes cylinders declared out of order to avoid to re-use them



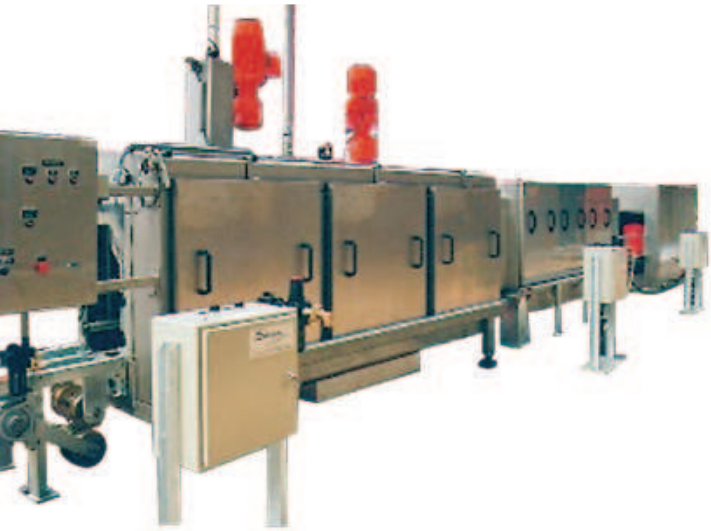
Press to flatten cylinders

- This press ensures complete destruction of the cylinders, by reducing their volumes for transport to recycling



Cylinder piercing unit

- Automatic on chain conveyor



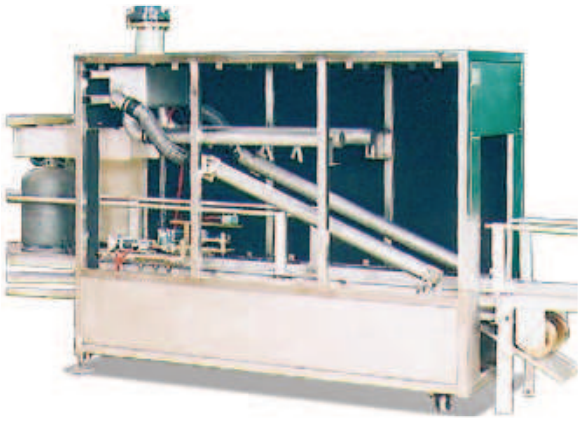
Washing tunnel

- With brushing, rinsing and drying tunnel



High pressure washing tunnel

- Capacity up to 1,800 cylinders per hour



Drying system



Pump skid for high pressure washing tunnel



Detergent spraying tunnel



Water decantation unit

- For high pressure washing tunnel

Siraga

GAS'LIB®

- 18 cylinders in an automatic vending as big as half of a parking space
- Payment by token, bar-code or credit card





DISTRI'GAS

- Extensible capacity from 24 to 48, 72 or 96 cylinders by adding moduls
- Energetic autonomy (solar panels)



VENDIS®

- Extensible capacity from 84 to 168 cylinders by adding a modul
- Multi-brand





CAGES

ENGINEERING & CONSULTING



Thanks to our Engineering branch, we provide turn-key industrial units, from design and engineering to complete manufacturing of industrial facilities, and so offer global and customised solutions that take into consideration all the characteristics and requirements of your project.







We achieve for you the following:

- LPG filling plants
- LPG cylinder manufacturing plants
- LPG cylinder maintenance plants
- Liquefied gas storages (pressurized, refrigerated or cryogenic)
- Hydrocarbon storages
- Oil and gas terminals
- Pumping stations on pipelines
- Hydrocarbon & gas treatment/separation plants
- Blending units and grease manufacturing plants
- Lubricant storage and conditioning plant
- Economic and feasability study
- Supply of LPG semi trailers
- Studies of natural gas transmission internal network

Our Engineering department also proceeds to the revamping of existing installations.





Within the achievement of your projects, our Engineering department offers the following range of services:

Consulting:
Feasibility and market studies, investment plan, project management

Management:
Control of all technical, economic and financial aspects of projects as well as respect of the provisional time schedule

Study:
Design and concept (FEED), detailed engineering: architectural, calculation of structure, piping, electricity, instrumentation, etc., equipment specifications, maintenance document, operating manuals

Procurement:
Equipment purchasing, forwarding

Civil works:
Subcontractor's contract establishment, performance of the works, coordination and supervision

Erection works:
Subcontractor's contract establishment, performance of the works, coordination and supervision

Technical assistance:
Technical assistance is provided by specialists in the concerned field and takes place during the first months after the commissioning of the installations.

Contact: engineering@siraga.com

A FEW STUDIES THAT WE HAVE CONDUCTED FOR SOME OF OUR CUSTOMERS:

- Underground LPG storage
- Technical & economical study for the replacement of side-entry valves by compact valves
- Feasibility, installation & maintenance of 450-liter LPG cylinders
- RFID tag on LPG cylinders
- Comparative study between steel and composite cylinders (with and without liner)
- Reduction of energy consumption in a filling plant



Siraga

LPG Academy®



OUR COMPLETE AND TAILORED TRAINING OFFER

You are attentive to the long-term, potential development of your employees, you wish them to adapt to the technical evolutions of your filling plant, you also wish them to be more qualified and thus more efficient ? Then discover or re-discover our complete and tailored training offer: the LPG ACADEMY !

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A tailored organization

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WHY CHOOSE THE LPG ACADEMY ?

For the exchange of expertise and know-how

- Because during these courses, you will alternate theory and practical exercises and be in contact with various LPG experts,
- Because the diversity of participants and experiences will enable to exchange ideas over your problems and to enhance your network.

For the professionalism

- Because the LPG Academy is a training organism recognized and integrated into the ISO certification field, to continuously ensure and improve your satisfaction,
- Because the internal trainers know the machines perfectly, because they have often participated in their conception, because they are experts in their field or also because the outside partners have been carefully selected since the creation of the LPG Academy. It represents the insurance of a training of quality.

For the services

- Because we fully organize your stay, from your arrival to the remittance of your diploma at the end of the course,
- Because after the work, it is time for relaxation. You will enjoy the French art of living to its fullest in a friendly and convivial atmosphere.

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Siraga ensures the installation & commissioning of the equipment on your production sites. Thanks to a team of local or multilingual technicians and experienced installers, Siraga guarantees a smooth installation process and provides a high-quality service.

Contact: cha@siraga.com



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Siraga markets a wide range of spare parts: with more than 50 000 spare parts available within the shortest possible time, at a competitive price. Thanks to a highly-experienced team, Siraga provides technical support and a service that exceeds your expectations.

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